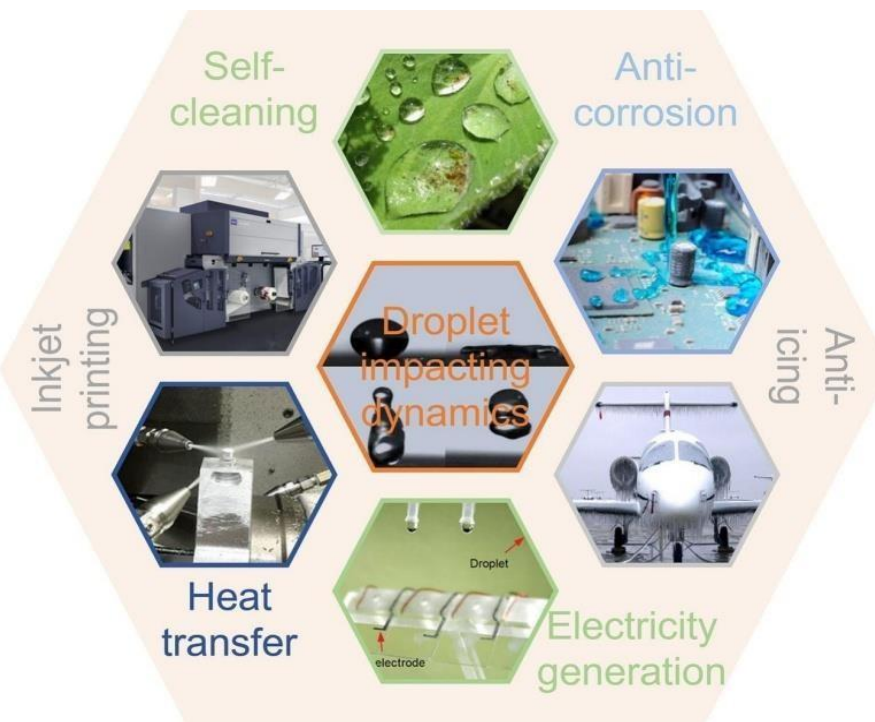


ABSTRACT

- Droplet impact on soft solids is important in many applications, including inkjet printing, spray coating, biomedical systems, and agricultural spraying
- Although droplet impact on compliant surfaces has been extensively studied for Newtonian liquids, the influence of fluid elasticity remains largely unexplored.
- We experimentally investigate the impact dynamics of viscosity-matched Newtonian and viscoelastic droplets on soft PDMS substrates with varying stiffness over a wide range of Weber numbers.
- High-speed imaging reveals that viscoelastic droplets exhibit significantly different spreading, recoiling, and splashing behavior compared with Newtonian droplets of the same viscosity.

APPLICATIONS OF DROPLET IMPACT



EXPERIMENTAL SETUP



Fig 1. Experimental set up



Fig 2. Rheometer

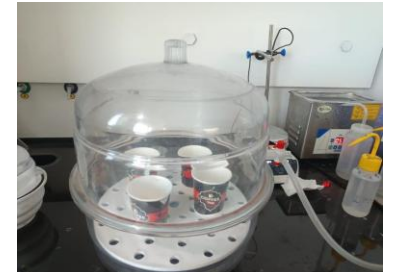
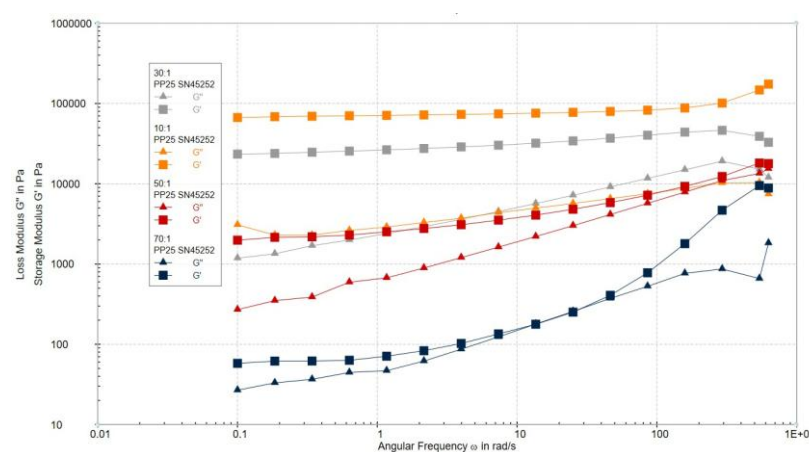


Fig 3. Desiccator

METHODS AND MATERIALS



Storage modulus increases with increase in cross linker, it means 10:1 is hardest and 70:1 is the softest substrate.

Fig 7: Loss modulus and storage modulus of different substrate

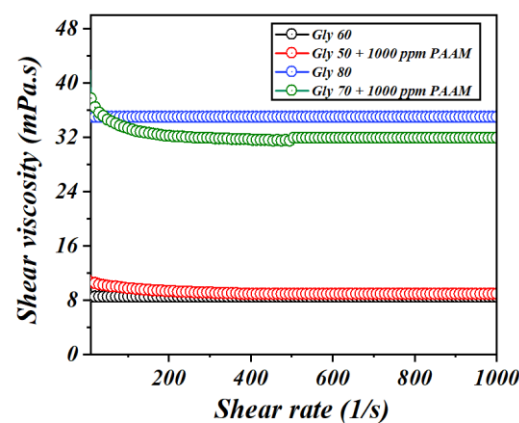


Fig 6: Shear viscosity of different fluids

Fluids	μ (mPa.s)
Gly 60 %	9
Gly 50% + 1000ppm	8.5
Gly 80%	35
Gly 70% + 1000 ppm	32

OBSERVATIONS

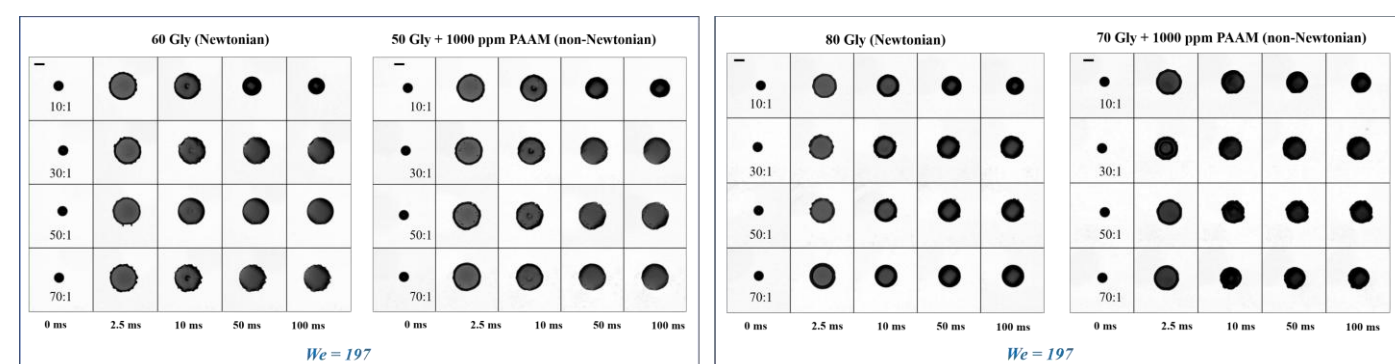


Fig 4: Comparing the effect of Newtonian and non Newtonian solutions on different soft solid substrates at $We = 197$, Diameter of the droplet (d) = 2.8 mm = scale bar length

- At $We = 197$, both Newtonian and viscoelastic fluids show almost same behavior for all substrate.

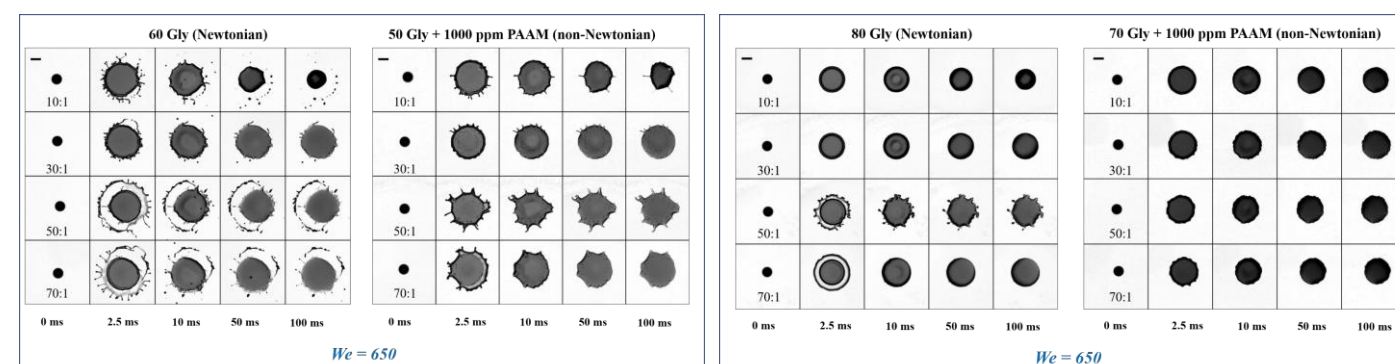


Fig 5: Comparing the effect of Newtonian and non Newtonian solutions on different soft solid substrates at $We = 650$, Diameter of the droplet (d) = 2.8 mm = scale bar length

- At $We = 650$, Newtonian droplet shows splashing while non Newtonian droplet suppress it due to elasticity.

Variation of maximum spread diameter (β_{max})

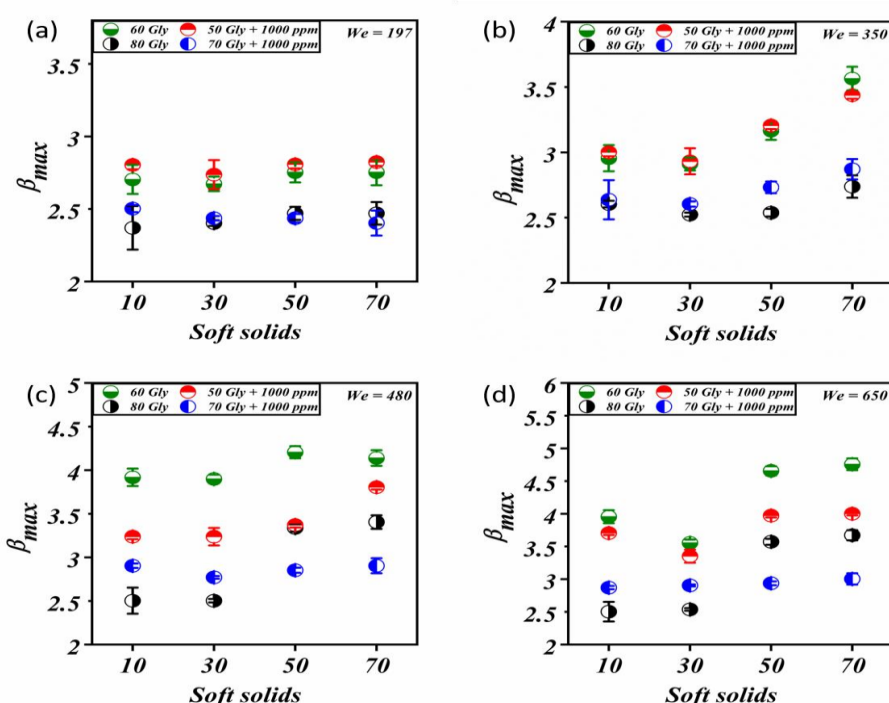


Fig 8: Variation of non dimensional maximum spread diameter vs soft solid softness for different fluids at fixed We (a) $We = 197$ (b) $We = 350$ (c) $We = 480$ (d) $We = 650$.

- At low We , max spread diameter is independent of softness and fluids.
- For a fixed substrate and fluid viscosity, max spread diameter increases with We .
- For a fixed We Retraction diameter increases with increases in softness of substrate for all fluids.

Variation of Retraction diameter (α)

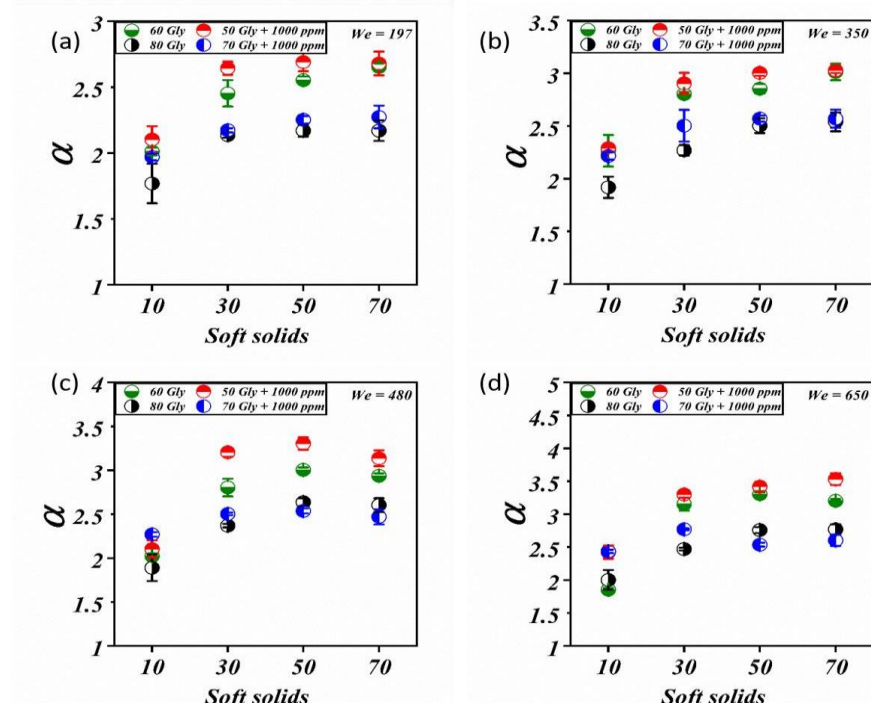


Fig 9: Variation of non dimensional retraction diameter vs soft solid softness for different fluids at fixed We (a) $We = 197$ (b) $We = 350$ (c) $We = 480$ (d) $We = 650$.

CONCLUSION

- At $We = 197$, both Newtonian (60% and 80% glycerol) and non-Newtonian (50% glycerol + 1000 ppm PAAM and 70% glycerol + 1000 ppm PAAM) droplets showed similar impact behavior on all substrates (10:1, 30:1, 50:1, and 70:1).
- At $We = 650$, Newtonian droplets exhibited splashing, whereas non-Newtonian droplets suppressed splashing due to viscoelastic effects and higher energy dissipation.
- Rheological analysis showed that the storage modulus increased with cross linker concentration, making the 10:1 substrate the hardest and the 70:1 substrate the softest. Substrate stiffness significantly affected droplet deformation and spreading dynamics.

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